「Prerequisites for Canonical Form.

「Def. Let $T: V \to V$ be a linear operator on V.

A subspace $W \leq V$ is called T-invariant if: $T[W] \leq W$.

Note: $\mathrm{Im}\, T$, $\ker T$, E_λ are all T-invariant.

「Lemma. A T-cyclic subspace generated by $x \in V$

$$W = \langle T^n(x) \mid n = 0, 1, 2, \cdots \rangle$$

is the smallest T-invariant containing $x \in V$.

Prf. T-invariant is clear. Suppose that W' is a T-invariant subspace containing $x \in V$.

Then, $\quad x \in W' \Rightarrow T(x) \in W' \Rightarrow T^2(x) \in W' \cdots$

By induction, for all $n \in \mathbb{N}$, $T^n(x) \in W'$. That is, W' contains generating set of W.

「Lemma. If W is T-invariant, then also $(T - cI)$-invariant, for any $c \in F$.

Proof. Let $x \in W$ be given. Then,

$$(T - cI)(x) = \underbrace{T(x)}_{\in W} - \underbrace{cx}_{\in W} \in W$$

Canonical Form.

Def. Let $T: V \to V$ be a linear operator, and $\lambda \in F$ be a scalar.
A non-zero vector $x \in V$ is called Generalized eigenvector of T corresponding to λ if;

$$\text{for some } p \in \mathbb{N}, \ (T-\lambda I)^p(x) = 0$$

Denote a set of generalized eigenvectors of λ as K_λ $\left(K_\lambda = \bigcup_{n=1}^{\infty} \ker(T-\lambda I)^n\right)$

Note: If a scalar $\lambda \in F$ has a generalized eigenvector s.t $(T-\lambda I)^p(x)=0$, then λ is eigenvector of T, Corresponding the eigenvector $(T-\lambda I)^{p-1}(x)$.

Thm. Let $T: V \to V$ be a linear operator, and $\lambda \in F$ be an eigenvalue of T.

1). Generalized Eigenspace K_λ is T-invariant subspace which contains E_λ.
2). For any eigenvalue $\mu \in F$ with $\mu \neq \lambda$, $K_\mu \cap K_\lambda = \{0\}$.
3). For any scalar $\gamma \in F$ with $\gamma \neq \lambda$, the restriction

$$(T-\lambda I)|_{K_\lambda} : K_\lambda \to K_\lambda ; x \mapsto (T-\lambda I)(x)$$

is bijection (That is, Isomorphism).

Proof. Clearly, $0 \in K_\lambda$. To use subspace criterion, let $c \in F$, $x, y \in K_\lambda$ be given.
That is, for some $p, q \in \mathbb{N}$, $(T-\lambda I)^p(x) = (T-\lambda I)^q(x) = 0$. Now,

$$(T-\lambda I)^{p+q}(cx+y) = c\cdot(T-\lambda I)^{p+q}(x) + (T-\lambda I)^{p+q}(y)$$
$$= c\cdot(T-\lambda I)^q(0) + (T-\lambda I)^p(0)$$

This implies $cx+y \in K_\lambda$.

To show that T-invariant, let $x \in K_\lambda$ s.t $(T-\lambda I)^p(x) = 0$. Then,

$$(T-\lambda I)^p(T(x)) = T \circ (T-\lambda I)^p(x) = T(0) = 0.$$

This implies $T[K_\lambda] \leq K_\lambda$

Let $\mu \in F$ be an eigenvalue of T s.t $\mu \neq \lambda$. Chose $x \in K_\mu \cap K_\lambda$. Then, for some $p, q \in \mathbb{N}$,

$$(T-\lambda I)^p(x) = (T-\mu)^q(x) = 0.$$

In other words, for $f_\lambda(t) = (t-\lambda)^p$, $f_\mu(t) = (t-\mu)^q$, $f_\lambda(T)(x) = f_\mu(T)(x) = 0$.
Since $F[t]$ is PID, it is also Bezout domain, there exist $q_1(t), q_2(t) \in F[t]$ s.t

$$q_1(t) f_\lambda(t) + q_2(t) f_\mu(t) = \gcd(f_\lambda(t), f_\mu(t)) = 1.$$

Therefore, $q_1(T) f_\lambda(T)(x) + q_2(T) f_\mu(T)(x) = q_1(T)(0) + q_2(T)(0) = 0 = x$.

Let $\gamma \in F$ with $\gamma \neq \lambda$ be given. Since K_λ is T-invariant, for any $x \in K_\lambda$,

$$(T - \gamma I)(x) = T(x) - \gamma \cdot x \in K_\lambda$$

That is, K_λ is $(T - \gamma I)$-invariant.

Let $x \in K_\lambda$, and suppose $(T - \gamma I)(x) = 0$. Then, $T(x) = \gamma x$, thus $x \in E_\gamma \subseteq K_\gamma$

But by 2., $K_\gamma \cap K_\lambda = \{0\}$, $x = 0$. That is, $\ker(T - \gamma I)|_{K_\lambda} = \{0\}$, thus 1-1.

Let $x \in K_\lambda$ be given. Put $p \in \mathbb{N}$ be the smallest Natural number s.t $(T - \lambda I)^p(x) = 0$.

Define

$$W = \langle \{x, (T - \lambda I)(x), \cdots, (T - \lambda I)^{p-1}(x)\} \rangle$$

Then, W equals to $(T - \lambda I)$-cyclic subspace generated by x,

thus W is the smallest $(T - \lambda I)$-invariant subspace containing x. thus $W \subseteq K_\lambda$.

Also, W is $(T - \gamma I)$-invariant. Now, a restriction

$$(T - \gamma I)|_W : W \to W ; y \mapsto (T - \gamma I)(y)$$

is well-defined injection, and surjection, since W is fin. dim.

Finally, there exists $y \in W$ s.t $(T - \gamma I)|_W(y) = x$, thus $(T - \gamma I)|_{K_\lambda}$ is surjection

「Thm. Let $T : V \to V$ be a linear operator on fin.dim V over F
If a charaterstic polynomial $\det(T-tI)$ splits over F,
then for an eigenvalue λ with multiplicity $m \geq 1$, the followings are hold;
1) $\dim K_\lambda \leq m$
2) $K_\lambda = \ker(T-\lambda I)^m$

Proof. Denote $f(t) = \det(T-tI)$ and $h(t) = \det(T_{K_\lambda} - tI)$.
If T_{K_λ} has another eigenvalue μ, then there exists a non-zero vector v s.t

$$v \in E_\mu \leq K_\mu \text{ and } v \in K_\lambda.$$

But, since $K_\mu \cap K_\lambda = \{0\}$ it is contradiction. Now,

$$h(t) = (-1)^d \cdot (t-\lambda)^d$$

where $d = \dim K_\lambda$. Since $h(t)$ divides $f(t)$, $\dim K_\lambda = d \leq m$.
Since $K_\lambda = \bigcup_{n=1}^{\infty} \ker(T-\lambda I)^n$, $\ker(T-\lambda I)^m \subseteq K_\lambda$ is trivial.
And, $f(t) = (t-\lambda)^m g(t)$ where $g(t)$ has no factor $(t-\lambda)$.
For any $\gamma \neq \lambda$, $(T-\gamma I)|_{K_\lambda}$ is bijection. Thus, $g(T)|_{K_\lambda}$ is bijection.
Let $x \in K_\lambda$. Then, for some $y \in K_\lambda$, $x = g(T)(y)$. Now,

$$(T-\lambda I)^m(x) = (T-\lambda I)^m(g(T)(y)) = f(T)(y) = 0.$$

by Cayley-Hamilton Theorem. Thus $x \in \ker(T-\lambda I)^m$.

「Thm. Let V be a fin. dim vector space, $T: V \to V$ be a linear operator.
If a characteristic polynomial $f(t)$ of T splits over F,
then for distinct eigenvalues $\lambda_1, \cdots, \lambda_k$ of T,

$$V = K_{\lambda_1} \oplus \cdots \oplus K_{\lambda_k}.$$

Proof. Denote $n = \dim V$, and

$$f(t) = (-1)^n (t-\lambda_1)^{n_1} \cdots (t-\lambda_k)^{n_k}.$$

And for each $1 \le j \le k$, define

$$f_j(t) = \prod_{\substack{i \neq j \\ 1 \le i \le k}} (t-\lambda_i)^{n_i}$$

Since f_j $(1 \le j \le k)$ are mutually disjoint, there exist $g_j(t)$ $(1 \le j \le k)$ such that

$$g_1(t) f_1(t) + \cdots + g_k(t) f_k(t) = 1.$$

Now, for any $x \in V$,

$$g_1(T) f_1(T)(x) + \cdots + g_k(T) f_k(T)(x) = x.$$

Meanwhile, for each $g_j(T) f_j(T)(x)$, the Cayley-Hamilton Thm gives

$$\begin{aligned}
&(T - \lambda_j I)^{n_j} \left(g_j(T) f_j(T)(x) \right) \\
&= (T - \lambda_j I)^{n_j} \left(f_j(T) g_j(T)(x) \right) \\
&= f(T) \left(g_j(T)(x) \right) \\
&= 0
\end{aligned}$$

That is, $g_j(T) f_j(T)(x) \in K_{\lambda_j}$. This proved $V = K_{\lambda_1} + \cdots + K_{\lambda_k}$.
To show the uniqueness, note that: since K_{λ_i} is T-invariant, $f_j(T)$-invariant.
Also, a restriction $f_j(T)|_{K_{\lambda_j}}$ is bijection, being composition of bijections.
Moreover, for $i \neq j$, $f_j(T)[K_{\lambda_i}] = f_j(T)\left[\ker (T - \lambda_i I)^{n_i}\right] = \{0\}$
Now, if $\sum_{i=1}^{k} v_i = \sum_{i=1}^{k} w_i$ where $v_i, w_i \in K_{\lambda_i}$ $(1 \le i \le k)$, then for each $1 \le j \le k$,

$$f_j(T)(v_j) = \sum_{i=1}^{k} f_j(T)(v_i) = \sum_{i=1}^{k} f_j(T)(w_i) = f_j(T)(w_j).$$

Since $f_j|_{K_{\lambda_j}}$ is injection, $v_j = w_j$.

Thm. Let V be a finite dimensional vector space, $T: V \to V$ be a linear operator.
If $\det(T-tI)$ splits over F as;

$$\det(T-tI) = (\lambda_1 - t)^{m_1}(\lambda_2 - t)^{m_2} \cdots (\lambda_k - t)^{m_k}.$$

Then, $V = K_{\lambda_1} \oplus \cdots \oplus K_{\lambda_k}$. Let β_{K_i} be a basis for K_{λ_i}.
The following statements are true;

1) $i \neq j \Rightarrow \beta_i \cap \beta_j$.
2) $\beta = \bigcup_{i=1}^{k} \beta_i$ is basis for V
3) for each $1 \leq i \leq k$, $\dim K_{\lambda_i} = m_i$.

Proof. 1) and 2) are clear. To show 3), see that;

$$\dim V = \sum_{i=1}^{k} m_i = \sum_{i=1}^{k} \dim(K_{\lambda_i})$$

And for all $1 \leq i \leq k$, $\dim K_{\lambda_i} \leq m_i$, thus $m_i = \dim K_{\lambda_i}$.

Def. Let $T: V \to V$ be a linear operator, and $x \in V$ be a generalized eigenvector of T corresponding to $\lambda \in F$. For the smallest $p \in \mathbb{N}$ s.t. $(T-\lambda I)^p(x) = 0$, a set

$$\{(T-\lambda I)^{p-1}(x), (T-\lambda I)^{p-2}(x), \cdots, (T-\lambda I)(x), x\}$$

is called cycle of x, and denote C_x. (Note: only $(T-\lambda I)^{p-1}(x)$ is eigenvector of $T-\lambda I$.)

Thm. Let $T: V \to V$ be a linear operator, $x \in V$ be a general. eigen. of T corresponding to $\lambda \in F$. For cycle C_x of x, $\langle C_x \rangle$ is T-invariant, and $[T|_{\langle C_x \rangle}]_{C_x}$ is Jordan block.

Proof. Denote $C_x = \{(T-\lambda I)^{p-1}(x), \cdots, x\}$.

For each $0 \le i \le p-1$,

$$(T-\lambda I)\left((T-\lambda I)^i(x)\right) = (T-\lambda I)^{i+1}(x) \in \langle C_x \rangle$$

Thus, $\langle C_x \rangle$ is $(T-\lambda I)$-invariant, so that T-invariant.

Meanwhile, $[T_{\langle C_x \rangle}]_{C_x} = [T-\lambda I|_{\langle C_x \rangle}]_{C_x} + [\lambda I_{\langle C_x \rangle}]_{C_x}$ and

$$[T-\lambda I|_{\langle C_x \rangle}]_{C_x} = \begin{bmatrix} 0 & 1 & 0 & & \\ 0 & 0 & 1 & & \\ \vdots & & 0 & \ddots & \\ \vdots & & & \ddots & 1 \\ 0 & & & & 0 \end{bmatrix}$$

gives the result.

Note: $C_x = \{(T-\lambda I)^{p-1}(x), \cdots, x\}$ are distinct; because if $0 \le i < j < p-1$ satisfies

$$(T-\lambda I)^i = (T-\lambda I)^j$$

$$\Rightarrow (T-\lambda I)^{\underbrace{p-j-1+i}_{=\, p-1-(j-i)}} = (T-\lambda I)^{p-1} = 0$$

Contradiction.

Thm. Let $\lambda \in F$ be an eigenvalue of $T: V \to V$.
Let $x_1, \dots, x_q \in K_\lambda$ be distincts vectors.
If initial vector of each C_{x_i} are mutually different and
initial vectors are linearly independent
then C_{x_i} are mutually disjoint and $C = \bigcup_{i=1}^{q} C_{x_i}$ is linearly independent.

Proof Denote that

$$C_x = \{(T-\lambda I)^{a-1}(x), \dots, x\}, \quad C_y = \{(T-\lambda I)^{b-1}(y), \dots, y\} \quad a \le b$$

If for some $1 \le i \le a$ $1 \le j \le b$

$$(T-\lambda I)^{a-i}(x) = (T-\lambda I)^{b-j}(y),$$

Then $(T-\lambda I)^{a}(x) = (T-\lambda I)^{b-j+i}(y) = 0 = (T-\lambda I)^{a-i+j}(x)$

If $i-j=0$, then these have same initial vector, contradiction.
If $|i-j|>0$, then contradiction to minimality.

$$\begin{pmatrix} a & & & \\ & b & & 0 \\ & & c & \\ & 0 & & \end{pmatrix}$$

$$\begin{pmatrix} a & 1 & & & \\ & b & 1 & 0 & \\ & & c & 1 & \\ & 0 & & \ddots & \\ & & & & \ddots \end{pmatrix}$$

Let $T: V \to V$ be a linear operator, and $f(T)$ be a characteristic polynomial of T.

Def. Suppose that $\lambda \in F$ be an eigenvalue of T. Define a set:

$$K_\lambda := \{x \in V \mid (T-\lambda I)^p(x) = 0 \text{ for some } p \in \mathbb{N}\} = \bigcup_{n=1}^{\infty} \ker(T-\lambda I)^n$$

It is called a generalized eigenspace corresponding to λ.

For non-zero $x \in K_\lambda$, the smallest positive integer $p \in \mathbb{Z}$ s.t. $(T-\lambda I)^p(x) = 0$ is called period of x. (or length)

(If $x=0$, period of x is assigned as zero.)

Lemma. Let $\lambda \in F$ be an eigenvalue of T. Then, the following properties are satisfied:

1). K_λ is T-invariant subspace of V containing E_λ.

2). If $\mu \in F$ is an eigenvalue of T s.t. $\mu \neq \lambda$, then $K_\mu \cap K_\lambda = \{0\}$

3) For any $c \in F$ s.t. $c \neq \lambda$, $T - cI|_{K_\lambda}$ is a bijection.

proof. Let $x, y \in K_\lambda$ and $\alpha \in F$ be given. Denote $p, q \in \mathbb{Z}$ as period of x, y, respectively.

Then,

$$(T-\lambda I)^{p+q}(\alpha x + y) = \alpha (T-\lambda I)^{p+q}(x) + (T-\lambda I)^{p+q}(y) = 0.$$

Clearly, $0 \in K_\lambda$. Therefore, subspace criterion gives $K_\lambda \leq V$.

To show T-invariant, let $x \in K_\lambda$. Then, for period $p \in \mathbb{Z}$ of x,

$$(T-\lambda I)^p(T(x)) = ((T-\lambda I)^p \circ T)(x) = (T \circ (T-\lambda I)^p)(x) = T(0) = 0.$$

Thus $T(x) \in K_\lambda$. Including E_λ is clear.

Show 2). Let $w \in K_\mu \cap K_\lambda$ be given. Let $p, q \in \mathbb{Z}$ be period of w for μ and λ, respectively.

Since $\mu \neq \lambda$ $(t-\mu)^p, (t-\lambda)^q \in F[t]$ are disjoint. Now, there exist $g_1(t), g_2(t) \in F[t]$ s.t.

$$g_1(t)(t-\mu)^p + g_2(t)(t-\lambda)^q = 1.$$

because $F[t]$ is P.I.D. Now,

$$(g_1(T)(T-\mu)^p + g_2(T)(T-\lambda)^q)(w) = 0 = w$$

Thus $K_\mu \cap K_\lambda = \{0\}$.

Finally, show 3). Since for any $w \in K_\lambda$, $(T-cI)(w) = 0 \Rightarrow w \in E_c \subseteq K_c$

By 2), $w=0$. That is, kernel of $(T-cI)|_{K_\lambda}$ is zero space, thus injective.

To show surjective, let $x \in K_\lambda$. For period $p \in \mathbb{Z}$ of x,

$$W = \{x, (T-\lambda I)(x), \cdots, (T-\lambda I)^{p-1}(x)\} \subseteq K_\lambda$$

is $(T-\mu I)$-invariant. Now, $(T-\mu I)|_W$ is injection on fin. dim space, thus surjection.

Now, we can find $y \in W$ s.t. $(T-\mu I)(y) = x$.

Thm. If $f(t)$ splits over F, then for any eigenvalue λ with multiplicity $m \in \mathbb{N}$,

$\dim K_\lambda \leq m$ and $K_\lambda = \ker\left((T-\lambda I)^m\right)$

proof. Let $h(t)$ be a characteristic polynomial of a restriction $T|_{K_\lambda}$.

Since for any distinct eigenvalue $\mu \in F$, $K_\mu \cap K_\lambda = \{0\}$,

λ is a unique eigenvalue of $T|_{K_\lambda}$. Moreover, since $h(t)$ divides $f(t)$,

$$h(t) = (\lambda - t)^d \quad (d \leq m).$$

Clearly, $\ker(T-\lambda I)^m \subseteq K_\lambda$.

Meanwhile, by assumption, $f(t) = (t-\lambda)^m \cdot g(t)$ where $g(\lambda) \neq 0$.

Therefore, $g(T)|_{K_\lambda}$ is bijection. Now, for any $x \in K_\lambda$, there is $y \in K_\lambda$ s.t. $g(T)(y) = x$

The cayley-Hamilton Thm gives

$$(T-\lambda I)^m(x) = (T-\lambda I)^m(g(T)(y)) = f(T)(y) = T_0(y) = 0.$$

Thus $x \in \ker(T-\lambda I)^m$.

$$\operatorname{Im} U = (T-\lambda I)|_{K_\lambda}[K_\lambda]$$

$$= (T-\lambda I)|_{K_\lambda}\left[\bigcup_{n=1}^{\infty} \ker (T-\lambda I)^n\right]$$

$$= \bigcup_{n=1}^{\infty} (T-\lambda I)\left[\ker (T-\lambda I)^n\right]$$

$T: V \to V$

$$V \supseteq K_\lambda \supseteq \operatorname{Im} U$$

$T|_{\operatorname{Im} U}$